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CompSci 161 Greedy Diagnostic Exam, Fall 2025

DO NOT OPEN EXAM UNTIL INSTRUCTED TO DO SO

SILENCE AND STORE ALL ELECTRONICS

This is a diagnostic exam intended to help you evaluate your readiness for the real exam.

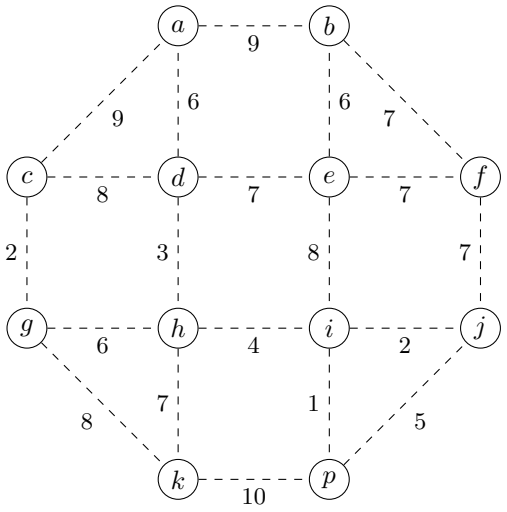
The following rules apply to you, whether you think they do or not. Read and understand them; failure to abide by these rules, or directions given by course staff during the exam, may result in disciplinary action, including but not limited to a failing grade in the class.

- This exam is solely for students enrolled in this lecture. Anyone not enrolled in this lecture may not take an exam.
- You may not open the exam or begin writing until the instructor has explicitly given you permission to do so.
- Keep acceptable identification readily accessible during the test. Proctors may request to see it.
- This exam is closed book, closed notes, and is individual effort. Once course staff begin passing out exams, you may not communicate with anyone other than proctors for any reason, nor may you have electronics, including calculators watches and phones, available to you during the test for any reason. **YOU DO NOT NEED A CALCULATOR!**
- If you leave your seat during the test for any reason, your instructor may collect it and deem you to have turned it in. Do not ask proctors for an exemption to this, they are not authorized to grant such.
- You must take the exam in your assigned seat unless the professor (not a TA) tells you otherwise. The instructor will call to cease writing at 6:45 PM, at which point you must immediately cease writing and close the exam. You may not write any further at that point, including finishing one's current sentence.
- If you believe a question is ambiguous, write at least two reasonable interpretations and indicate clearly which one you will be using. Then answer your question with that assumption. Unless your interpretation makes the problem much more trivial than intended, we will grade your response as if one of us had made that clarification.
- The purpose of the real exam is to evaluate how well *you* understand the material presented in the course. It is an academic integrity violation to do anything that subverts the goals of this assessment including, but not limited to, not doing your own work or submitting that of anyone else.
- We will only grade responses marked in the space provided for each question.

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Nothing you write on this page will be graded. The next page in this booklet contains a spot to answer these questions. You may use this page as scratch paper if you would like, and room to do so exists.

1. (2 points) The following question concerns computing a Minimum Spanning Tree for the following graph.



For each of the following edges, is it part of *every* correct minimum spanning tree, *no* minimum spanning trees, or *some* (but not all) minimum spanning trees?

(a) (i, j)

(b) (p, j)

(c) (b, f)

(d) (a, d)

2. (2 points) Use Huffman's algorithm to produce an encoding tree for the following character distribution. Note that there will be ten leaf nodes. You may break ties however you would like.

The count of characters is:

A	L	G	O	R	I	T	H	M	S
6	8	2	3	1	10	2	1	8	7

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For question 1, **circle** the correct answer for each part.

(a) every no some

(b) every no some

(c) every no some

(d) every no some

Draw your tree for question two below

Nothing you write on this page will be graded. The next page in this booklet contains a spot to answer these questions. You may use this page as scratch paper if you would like, and room to do so exists.

3. (2 points) Consider the problem you might be faced with if you are in charge of the dunk tanks at next year's ICS Week. The dunk tank is a fun event: a professor sits on a chair above a large container of water. A student throws a ball at a target, and if they hit the target, the professor falls into the water. In 2025, one professor agreed to be on the dunk tank during ICS week; inspired by that example, we have n students who have volunteered to help at ICS week next year. Each student worker has signed up for exactly one shift; student i will arrive at time a_i and finish their shift at time f_i. You may assume that, at every point in time, from the first arrival to the last departure, there is at least one student on the schedule at that time.

   However, management realizes students would rather watch the professor at the dunk tank than work, but we need to make sure the booth at the front has a person there at all times. As such, our goal is to select *the fewest number of student volunteers* to remain at the booth, such that every moment of the event has at least one student worker present. Everyone else can go watch the dunk tank.

   Prove that the following **greedy algorithm** will correctly do so: of all students that start at the start of the event (earliest of all $\{a_i\}$), take the one whose shift ends last. For the second (and future) selections, choose the latest ending time out of those that start no later than the finish time of the previous selection.

Remember to write your answer on the answer page →, not here, although you may use this space for scratch work.

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Write your proof for question three on this page. It is possible that you won't use the entire page.

Scratch page. Nothing you write on this side of the page will be graded.

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